SHOP.PY

Generative Engine Optimization Analytics Platform

5-Page Business & Marketing Plan

2026 HBCU Battle of the Brains Case Competition
"The New Front Door: Trustworthy AI Product Discovery"

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Executive Summary

AI assistants are rapidly becoming the primary discovery layer for consumer purchasing. Shoppers increasingly ask questions like *"What is the best laptop under $500?"* and receive ranked answers directly in the AI, often bypassing brand websites. This shift creates a critical opportunity for brands to regain visibility and influence in AI driven commerce. Shop.py is a B2B SaaS analytics platform designed to help companies track, measure, and optimize how their products appear in AI-generated shopping results through Generative Engine Optimization (GEO)**.** GEO represents an emerging category within the broader digital marketing analytics and SEO software markets. Global SEO and search analytics tools currently represent a multibillion-dollar industry, and as product discovery shifts from search engines to AI assistants, GEO analytics is expected to capture a growing share of this spend. Our platform enables brands to monitor ranking positions, sentiment signals, competitive comparisons, and recommendation drivers, allowing actionable optimization of product content, reviews, and pricing signals. Shop.py's intelligence is powered by ShopSmart, a free consumer browser extension that enhances AI shopping sessions with coupon codes, aggregated product insights, and social video content from social media platforms like TikTok and Instagram, while capturing anonymized, consented behavioral data from real shopping interactions. This creates a proprietary data flywheel no competitor can replicate. The GEO market is currently valued at $886 million (2024) and is projected to reach $7.32 billion by 2031 (34% CAGR), highlighting the urgency for brands to act. Just as SEO became essential infrastructure for search-driven commerce, GEO analytics will become essential infrastructure for AI-driven product discovery. We recommend that companies adopt Shop.py to gain early insights into AI-driven product discovery, leverage behavioral data to optimize listings and marketing strategies, and expand beyond our initial focus on technology retailers, where 54% of consumers are shopping via AI platforms, into high-potential verticals such as home goods (44%), travel (43%), and health products (41%). By doing so, brands can improve their visibility in AI recommendations, drive sales growth, and build a sustainable competitive advantage in the rapidly expanding GEO market.

Problem	Solution	Advantage	Revenue Model
Brands are invisible in AI results with no tools to measure or act on it	GEO analytics dashboard + ethics hallucination monitoring powered by real consumer	Only platform using real AI shopping interactions vs. competitors' bot	Usage-based API pricing: pay per call, scaled by company size

	data	simulations	

Problem, Solution & Pricing

Traditional digital marketing signals, website traffic, click-through rates, and search rankings, no longer reflect true demand when purchase decisions are made inside a ChatGPT or Gemini session. PartnerCentric research shows 64% of consumers plan to use AI for technology purchases in 2026. AI systems hallucinate product details including pricing, features, and availability, spreading misinformation at the moment of purchase intent. Competitors Semrush, Agenxus, and GrowthX address adjacent problems but none offer dedicated GEO analytics powered by real consumer behavioral data.

Platform Components

ShopSmart Browser Extension: Free consumer tool surfacing coupon codes, aggregated reviews, and TikTok/Instagram video reviews inside AI shopping sessions. Users grant consent for anonymized behavioral data collection, powering the analytics platform with authentic query patterns, product categories, and AI platform usage data that bot-based competitors cannot access. Aggregated behavioral data is normalized using query clustering and statistical sampling techniques to ensure that insights reflect broader AI shopping patterns rather than individual user behavior.

GEO Analytics Dashboard: B2B SaaS platform translating ShopSmart data into brand intelligence. Metrics include citation frequency and position, entity coverage, impression share versus competitors, and ranking stability. Tier 2 adds optimization recommendations, competitive gap analysis, predictive modeling, and cloud-based historical storage.

Ethics Monitoring Layer: Automated hallucination detection comparing AI-generated product claims against verified client catalogs. Discrepancies are severity-scored (low, medium, critical) and surfaced with remediation guidance. Monthly ethics dashboards measure whether corrections are reducing AI misinformation over time. The system does not directly modify AI model outputs. Instead, it alerts brands when inaccurate product claims appear in AI-generated responses, enabling companies to correct the underlying sources that influence AI knowledge systems such as product pages, structured data, and third-party review platforms.

Usage-Based API Pricing

Clients are billed per API call, meaning cost scales directly with data consumption and company size. A typical GEO analysis request may evaluate dozens of prompts across multiple AI platforms and product categories, resulting in thousands of API calls per brand per month. Each call triggers cloud data retrieval, AI prompt execution, and analytics computation, with a marginal infrastructure cost of $0.0008 to $0.0015 per call (AWS Lambda benchmarks). We apply a 10x to 15x SaaS margin, standard for data API products such as Twilio and Snowflake, validated by Bessemer Venture Partners' finding that usage-based SaaS achieves 15 to 20 percentage points higher net revenue retention than seat-based models.

Tier	Company Size	API Calls / Mo.	Per-Call Rate	Est. Annual Cost
Starter	SMB (<$10M rev.)	5,000	$0.012	$6K to $15K

Growth	Mid-Market ($10M to $100M)	25,000	$0.010	$30K to $75K
Enterprise	Enterprise ($100M+ rev.)	100,000+	$0.008	$150K to $300K

Marketing Plan

Shop.py's core revenue comes from enterprise clients purchasing access to the GEO analytics dashboard. Marketing strategy is therefore B2B-first, with the majority of budget and effort directed toward acquiring and retaining enterprise accounts. A secondary, smaller allocation supports ShopSmart consumer extension growth because extension adoption is what generates the behavioral data that makes the dashboard valuable. ShopSmart is a means to product excellence, not a parallel business.

Budget Allocation: B2B Dashboard vs. ShopSmart Extension

Marketing Channel	Focus	Year 1 Budget	Year 2 Budget	Goal
Enterprise Direct Sales	B2B Dashboard	$160,000 (27%)	$280,000 (28%)	15 to 20 signed accounts Year 1
Content & Thought Leadership	B2B Dashboard	$90,000 (15%)	$140,000 (14%)	Inbound pipeline from GEO benchmark reports
Conference & Events	B2B Dashboard	$80,000 (13%)	$120,000 (12%)	Brand credibility at Shoptalk, MarTech, NRF
Agency Partner Program	B2B Dashboard	$70,000 (12%)	$130,000 (13%)	Warm intros via digital marketing agencies
Paid B2B Digital (LinkedIn)	B2B Dashboard	$60,000 (10%)	$90,000 (9%)	Targeted ads to CMO, VP Marketing, Dir. eCommerce
ShopSmart Extension Growth	Consumer Extension	$90,000 (15%)	$150,000 (15%)	50,000 users Year 1 to power data pipeline
Brand & PR	Both	$50,000 (8%)	$90,000 (9%)	Press coverage and category authority
TOTAL	**85% B2B / 15% B2C**	**$600,000**	**$1,000,000**	

Primary: B2B Dashboard Marketing (85% of Budget)

1. Enterprise Direct Sales

Shop.py's highest-priority channel is a dedicated outbound sales function targeting VP of Marketing, Director of eCommerce, and Chief Marketing Officer roles at technology retailers and SaaS companies with revenues above $50 million. Sales reps will use LinkedIn Sales Navigator for prospecting, personalized email sequences anchored in GEO benchmark data relevant to the prospect's category, and a 30-day free dashboard trial offer for qualified accounts. Year 1 goal is 15 to 20 signed contracts averaging $75,000 annually. A quota-carrying sales team of two reps in Year 1, scaling to five in Year 2, will drive this motion.

2. Content Marketing & Thought Leadership

Shop.py will establish category authority in GEO through a quarterly GEO Visibility Benchmark Report, the first of its kind, showing which brands are winning and losing AI recommendation share by industry. These reports will be gated behind an email capture form, generating a high-intent lead list of marketing leaders seeking to understand their AI standing. Supporting content includes a weekly blog covering GEO strategy, case studies from pilot clients showing measurable citation improvement, and a monthly webinar series on AI commerce trends co hosted with marketing agency partners. HubSpot's 2024 State of Marketing report confirms that B2B companies publishing consistent thought leadership generate 67% more leads per month than those that do not. **3. Conference & Event Marketing**

Shop.py will exhibit at Shoptalk (March), MarTech Conference (April), and NRF Nextech (January) annually. These events concentrate the exact buyer profile, retail and SaaS marketing executives, who control GEO analytics budgets. Booth presence will feature a live GEO visibility audit, where prospects can see their brand's real AI citation performance in real time, creating an immediate, visceral demonstration of the platform's value. Conference sponsorship of breakout sessions on AI commerce will reinforce category leadership.

4. Digital Marketing Agency Partner Program

Digital marketing agencies that serve technology retail clients are a high-value referral channel because they are already in conversation with the exact clients Shop.py needs. Shop.py will establish a tiered agency partner program offering revenue share of 10 to 15% on referred contracts, co-branded GEO audit reports agencies can deliver to their own clients, and white-label dashboard access for agencies offering GEO as a managed service. Year 1 target is 10 active agency partners generating a combined $300,000 in referred ARR. **5. Paid B2B Digital (LinkedIn)**

LinkedIn Sponsored Content and InMail campaigns will target marketing decision-makers by job title, industry, and company revenue range. Ad creative will lead with category education, specifically the finding that 54% of consumers now use AI for technology purchases, before presenting Shop.py as the visibility solution. Retargeting campaigns will nurture website visitors who downloaded benchmark reports but did not convert to trial. Target cost-per-lead of $120 to $180, consistent with LinkedIn B2B SaaS benchmarks per Demand Gen Report 2024.

Secondary: ShopSmart Consumer Extension Growth (15% of Budget)

ShopSmart is marketed primarily on its consumer value proposition: save money and get richer product information while shopping with AI. Budget is allocated to influencer partnerships with personal finance and tech review creators on YouTube and TikTok ($40,000), affiliate placements on deal communities such as Slickdeals and DealNews ($30,000), and Chrome Web Store optimization and search ads targeting AI shopping queries ($20,000). The goal is 50,000 active users by end of Year 1, sufficient to produce statistically meaningful GEO behavioral data across major product categories. Extension growth is not measured as a revenue line but as a data quality metric that directly improves the dashboard product.

Implementation Plan & Financials

Implementation Phases

Phase	Timeline	Key Actions	Milestone

1: Build	Months 1 to 3	Build ShopSmart MVP and GEO API infrastructure on AWS; configure anonymization pipeline; begin SOC 2 compliance process	Extension live; API handling 5K calls/mo.
2: Pilot	Months 4 to 6	Onboard 3 to 5 pilot clients at reduced rate; grow ShopSmart to 10,000 users; validate GEO metrics accuracy; activate hallucination detection pipeline	First paying clients; proof-of-concept dataset established
3: Launch	Months 7 to 12	Commercial launch of Starter and Growth tiers; publish inaugural GEO Benchmark Report; activate agency partner channel; scale to 50,000 ShopSmart users	15 to 20 clients; $1.5M ARR; 50K extension users; Revenue Tier Mix: Growth (75%), Enterprise (15%), Starter (10%)
4: Scale	Year 2	Launch Enterprise tier; expand into SaaS and home goods verticals; add predictive visibility modeling; grow ShopSmart to 200K users	50 to 60 clients; $5M ARR; Series A readiness
5: Expand	Years 3 to 4	International expansion; Salesforce and HubSpot integrations; multi-language AI monitoring	120 to 150 clients; $15M ARR

Financial Projections

Projections are anchored to three research benchmarks: Bessemer Venture Partners' 2024 finding of ~3x YoY ARR growth for early-stage SaaS with product-market fit; OpenView Partners' 2024 data showing usage-based SaaS achieves 115 to 130% net revenue retention as clients expand API usage; and the GEO market's 34% CAGR (Grand View Research 2024) validating the revenue trajectory. As seen in the Launch Phase ARR, early adoption is expected to skew toward the Growth tier as mid-market companies test GEO analytics before expanding into Enterprise usage levels.

Metric	Year 1	Year 2	Year 3	Research Basis
Paying Clients	15 to 20	50 to 60	120 to 150	~3x YoY growth (BVP 2024)
Avg. Contract Value	$75,000	$90,000	$110,000	API upsell via usage expansion (OpenView 2024)
Annual Recurring Revenue	$1.5M	$5M	$15M	Aligned to GEO market 34% CAGR
Operating Expenses	$2.8M	$4.2M	$8.5M	Eng. ($1.6M), Sales/Mktg ($600K), Infra ($300K), Legal ($200K), Ops ($100K)

Net Income / (Loss)	($1.3M)	$0.8M	$6.5M	Break-even Year 2; consistent with early SaaS benchmarks

Capital Stack & Funding Strategy

Source	Amount	Timing	Justification
HBCU-Focused Angels & Impact VCs	$500K	Pre-seed Month 1	Directly reachable through competition and HBCU alumni networks; active in early-stage tech
(HBCU.vc, Harlem Capital, Fearless Fund)			
NSF SBIR Phase I Grant (Non-Dilutive)	$500K	Pre-seed Months 3 to 6	NSF Phase I awards $275K; Phase II up to $1M for AI/data infrastructure. Ethics monitoring qualifies under responsible AI criteria
Seed VC Round (MarTech / AI Commerce Focus)	$1.5M	Pre-seed Month 4	Pitch: proprietary data moat, usage-based NRR, no dominant GEO player. Comparable seed rounds for analytics SaaS: $1M to $3M (Crunchbase 2024 to 2025)
Series A VC Round	$8M to $12M	Year 3	Triggered at $5M ARR and positive net income. Median Series A for analytics SaaS: $8M to $15M (PitchBook 2024). Funds international expansion and enterprise integrations

Operations, Risk & ESG

Team & Infrastructure

Core roles at launch: CEO (strategy and investor relations), CTO (platform and data infrastructure), Head of Data Science (analytics and ML pipeline), Head of Enterprise Sales (B2B acquisition), Head of Marketing (B2B growth and content), and Head of Ethics and Compliance (hallucination monitoring and regulatory alignment). Technical infrastructure runs on AWS for compute and storage and dbt for data transformation. SOC 2 Type II certification is targeted within Year 1 and is a prerequisite for enterprise sales at accounts above $100M revenue.

Risk Management

Risk	Likelihood	

AI platforms restrict front-end data access	Medium	Extension captures user-visible interactions, not API data. Monitor across ChatGPT, Gemini, and Perplexity simultaneously to reduce single-platform dependency.
Consumer privacy regulation tightening (GDPR, CCPA)	High	Collect only anonymized aggregate data with explicit opt in. SOC 2 from launch. Engage privacy counsel proactively before expansion into EU markets.
Slow enterprise client adoption in Year 1	Medium	Offer reduced-rate 90-day pilot contracts. Use GEO Benchmark Report data as proof of concept. Agency partner channel provides warm introductions to budget holders.
Large competitor (Semrush, Adobe) enters GEO space	Low to Medium	Proprietary behavioral dataset from ShopSmart and integrated ethics monitoring are not replicable without equivalent consumer distribution. Speed to market compounds advantage.

ESG Commitments

Environmental: ShopSmart video content loads on user interaction only and is never stored, minimizing cloud energy consumption.

Social: Usage-based pricing lets SMBs access GEO analytics proportional to their scale, democratizing AI commerce intelligence. All data collection is consent-based and minimized.

Governance: The ethics monitoring layer reduces AI-generated product misinformation across the broader ecosystem, protecting consumers and brands beyond Shop.py's direct client base.

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